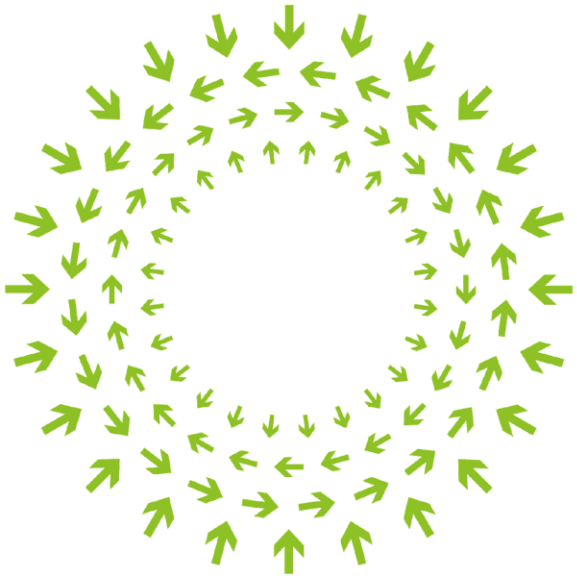


OPEN

Compute Project

M-PNP Interoperability Design Specification

Part of the

Modular Hardware Systems (MHS) Rev 1.x Family

Version 0.3.0

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Acknowledgements

With the hope of making this specification useful for the entire OCP community, we acknowledge and appreciate the contributions, review, and feedback of various individuals and companies that participated in MHS.

2. Version Table

Date	Version #	Description
9/12/23	0.01	Initial version
9/9/24	0.1	Removed sections. Cleaned up many sections. Addressed comments.
10/4/24	0.3.0	Addressed comments and removed sections to reduce scope for 0.3 release.

3. Scope

This document defines technical specifications for the Platform Interoperability Design Specification used in the MHS family.

Any supplier seeking OCP recognition for a hardware product based on this specification must be compliant with all mandatory features or requirements listed in the section entitled, “Specification Compliance Checklist”, of this document.

3.1. Vision Statement

The OCP MHS standards body defines interoperability for MHS supporting systems to require and use industry standard processes using accepted algorithms and protocols to discover, manage, optionally secure, and maintain systems based on MHS components. Interoperability means that all discoverable components will be interoperable in a compatible manner such that a DC-SCM or other controlling entity can perform all necessary functions for the management of the entire system. This is with the goal of having all components discoverable in a self-describing manner such that a DC-SCM can operate such components with little to no a-priori advanced knowledge to perform basic functions.

3.2. Interoperability Definition

Interoperability is the capability of a set of MHS compliant components to work seamlessly together mechanically, electrically, and ***logically***. Components must adhere to all specification requirements in support of those goals such that system software will be enabled to perform the necessary functions to discover, without a-priori knowledge, boot, and manage the components as a collective whole with defined methods and processes. Interoperability is achieved when devices from different vendors are mechanically, electrically, and logically compatible so that unusual (not following an existing standard) and/or proprietary integration steps are not required to operate to design goals and security constraints.

3.3. Specific Interoperability Specification Definitions

This section describes the levels of interoperability as it pertains to the entire system as a whole.

- **Plug and Code** – At this level, base specifications only cover electrical and mechanical interoperability. Interoperability between components above the electrical and mechanical domains are not provided by existing specifications thus compatibly is the responsibility of the system integrator. The system integrator is responsible for developing everything necessary to ensure that the system is functional.
- **Plug and Play ‘Lite’** – At this level, the interoperability framework is defined but there is a still a significant amount of code development work necessary to stitch together the framework pieces in order to get the system to a baseline level of functionality (e.g. booting to OS). This interoperability framework may be embodied in a set of required supporting specifications that outline commonly needed materials and processes such as component FRU information and a common discovery process.
- **Baseline Plug and Play** – At this level, it means that minimal code and configuration work would be necessary to describe the baseline functionality of a platform to achieve a basic level of functionality. The intention is that system integrators would only need to configure their system specific implementation via

standardized configuration files and/or other mechanisms (e.g. OpenBMC configuration files) and their BIOS firmware builds to describe the system. Any additional code would be for OEM extensions or enablement of OEM specific IP, modules, etc. In order to support the plug-and-play vision, all required interoperability specifications and additional supporting specifications will have been completed. A baseline reference implementation specification would have been provided which can be used to guide system integrators for their specific platform implementations. Additionally, OEM extension capabilities would be defined at this stage.

3.4. Typical OCP Sections Not Applicable

This is a design specification, requiring other MHS specifications to fully define a design. The following typical Sections of an OCP specification are not included because they are not applicable to this specification.

Rack Compatibility
Physical Spec
Rear Side Power, I/O, Expansion
Mechanical
Onboard Power System
Environmental Regulations/Requirements
Prescribed Materials

3.5. In-Scope

The specification defines component interoperability for a MHS supporting platform. This includes the HPM, SCM, Power Supplies & Distribution Boards, Risers / Backplanes, Option Cards, and additional components necessary to assemble a fully functional and operable platform.

Specification In-Scope List:

- Platform component requirements that support common interoperability
- Statements of interoperability as they pertain to supporting design specifications
- System management requirements not covered within existing required specifications
- Security requirements not covered within existing required specifications
- Non-functional requirements not covered within existing required specifications

3.6. Out of Scope

Defining electrical and mechanical interoperability is a required prerequisite for M-PNP interoperability and is already covered within the set of referenced documents. Thus it is out of scope to define it within this specification.

Specification Out-Scope List:

Operational Behavioral Definitions of system behaviors such as blink patterns, LED colors, Boot times.

3.7. MHS Family of Specifications

The Data Center – Modular Hardware System (MHS) Rev 1.0 family of specifications are written to enable interoperability between key elements of datacenter and enterprise infrastructure by providing consistent interfaces and form factors among modular building blocks. At the time of this publication there are the following specification workstreams:

- **M-FLW** (**M**odular Hardware System **F**ull **W**idth Specification) – Host Processor Module (HPM) form factor specification optimized for using the full width of a Standard EIA-310 Rack mountable server. The specification is not limited to use within the EIA-310 Rack but is used to serve as a template for a common target where the design is expected to be utilized.
- **M-DNO** (**M**odular Hardware System Partial Width **D**e**n**sity **O**ptimized Specification) – Host Processor Module (HPM) specification targeted to partial width (i.e., ½ width or ¾ width) form factors. Such form factors are often depth challenged and found not only in enterprise applications but also in Telecommunications, Cloud and Edge Deployments. While the EIA-310 Rack implementation is chosen as a key test case for use, the specification is not limited to use within the EIA-310 Rack but is used to serve as a template for a common target where the design is expected to be utilized.
- **M-CRPS** (**M**odular Hardware System **C**ommon **R**edundant **P**ower **S**upply Specification) – Specifies the power supply solutions and signaling expected to be utilized by MHS compatible systems.
- **M-PIC** (**M**odular Hardware System **P**latform **I**nfrastructure **C**onnectivity Specification) – Specifies common elements needed to interface a Host Processor Module (HPM) to the platform/chassis infrastructure elements/subsystems. Examples include power management, control panel and cooling amongst others.
- **M-XIO** (**M**odular Hardware System E**x**tensible **I**/**O**) – Specifies the high-speed connector hardware strategy. An M-XIO source connector enables entry and exit points between sources such as Motherboards, Host Processor Modules & RAID Controllers with peripheral subsystems such as PCIe risers, backplanes, etc. M-XIO includes the connector, high speed and management signal interface details and supported pinouts.
- **M-PESTI** (**M**odular Hardware System **Pe**ripheral **S**ideband **T**unneling **I**nterface) – Specifies a standard method for discovery of subsystems, self-describing attributes, and status (i.e., versus a priori knowledge/hard coding firmware and BIOS for fixed/limited configurations). Examples: vendor/module class, physical connectivity descriptions, add-in card presence, precise source to destination cable coupling determination.
- **M-SIF** (**M**odular Hardware System **S**hared **I**n**f**rastructure Specification) – Specifies data and management plane interfaces along with power and cooling connections that are expected to be present within a modular platform utilizing shared elements.
- **M-PNP.FRU_Discovery_Boot -** Specifies management plane collaterals with respect to common discovery and management of HPMs and peripheral subsystems. Examples include this FRU-related document, HPM FPGA guidelines, schemas and templates for HW descriptions that ease FW/BIOS platform enablement for new HPMs, peripherals and systems.
- **M-PNP.FPGA -** Specifies HPM FPGA requirements and guidelines for common BMC interfacing and descriptive information.

- **DC-SCM 2.1** (Data Center Secure Control Module Specification) - Specification for a control module containing the system BMC. This module implements a Data Center Secure Control Interface (DC-SCI) used for interfacing with a Host Processor Module (HPM).
- **HPM Common Circuits Type 1**- Specifies the set of HPM common circuits as interfaced by the SCM and it's BMC as categorized into types. Type 1 refers to a common single node platform.

To access additional MHS specifications, please visit the OCP Server WIKI.

3.8. Related OCP Specifications

The list below contains the set of related OCP specifications referenced in this document.

- **OCP NIC 3.0 Design Specification -** Specifies a standard for Network Interface Cards (NIC)s.

3.9. Terminology

Table 1: Terminology

Standardized Term	**Meaning**	**Alternative Terms**
+12V	12V Main Power from M-CRPS	
+12VStby	12V Standby from M-CRPS	
ACPI	Advanced Configuration and Power Interface	
BMC	Baseboard Management Controller	
CEM	Card Electromechanical Specification	
MHS	Data Center - Modular Hardware System	
DC-SCM	Data Center - Secure Control Module	
Flex I/O	Flexible I/O	
FPGA	Field-Programmable Gate Array	CPLD - Complex Programmable Logic Device
FRU	Field Replacement Unit: . The widget that can be individually services in the field . A FRU Information Device (EEPROM) . FRU Information (Data format such as IPMI Information Storage or PLDM FRU)	VPD (Vital Product Data) - Typically avoided due to PCIe Base Spec's different definition/use
HPM	Host Processor Module PCB or PCBA form factor defined in M-FLW or M-DNO	
M-CRPS	Modular Hardware System Common Redundant Power Supply	PSU, CRPS

MCU	A microcontroller unit	uC
M-DNO	Modular Hardware System Partial Width Density Optimized HPM Form Factor	DNO
M-FLW	Modular Hardware System Full Width HPM Form Factor	FLW
MGMT	Management	Sys Mgmt
M-XIO	Modular Hardware System eXtensible I/O	XIO
P-Card	Peripheral Card (such as a riser, backplane or interposer)	
PCB	Printed Circuit Board	PWB (Printed Wiring Board)
PCBA	Printed Circuit Board Assembly	PBA, PWA (Printed Wiring Assembly)
PCP	Primary Control Panel	RCP (Right control panel from front view)
PDB	Power Distribution Board	
PIB	Power Interface Board	
PICPWR	Platform Infrastructure Connectivity Power Distribution Connector	
PMBus	Power Management Bus	PMB
SCP	Secondary Control Panel	LCP (Left control panel from front view)
RoT	Root of Trust	
RFU	Reserved for Future Use	RSVD
SGPIO	Serial GPIO	
Smart NIC	A programmable Network Interface Card used to improve data center networking performance, security, features, and flexibility.	IPU or DPU
TDP	Thermal Design Point	

3.9.1. Industry Standards Terminology

The table below contains industry standards terminology.

TABLE 1: STANDARDS TERMINOLOGY

Standardized Term	Meaning	Alternative Terms
Shall	Indicates a requirement or something that must be done.	
May	Used to indicate that something is permitted.	
Should	Indicates a recommendation, which is optional but still a good idea.	

Can	Used to indicate that something is possible, for example, that an organization or individual is able to do something	

4. MHS Common Component Hardware Interoperability Requirements

This section outlines the common components specified for use by all platforms supporting MHS form factors.

4.1. Components Overview

The table below defines the set of MHS components from the perspective of how they are discovered and what they contain. In the context of this table, we define the term 'active' as having a programmable device that executes programmable logic or code, and 'passive' as not containing a programmable device with executable logic or code.

Component	Programmable Device*	FRU Information
DC-SCM	Shall Have	Shall Have**
HPM	Shall Have	Shall Have
M-CRPS	Shall Have	Shall Have
Active Backplane / Riser / Extension Board	May Have	Shall Have
Passive Backplane / Riser / Extension Board	Shall Not Have	Shall Have
Option Card (PCIe)	May Have	Shall Have
Passive Utility Boards	Shall Not Have	Shall Have
Active Intelligent Fans and Cooling Components	May Have	Shall Have
Direct Connect Fans and Cooling Components	Shall Not Have	May Have

*Note: Programmable device includes micro-controllers, micro-processors, FPGAs / CPLDs, and anything that supports executable logic or code.

**Note: DC-SCM specification does not explicitly indicate that it requires a FRU EEPROM or EEPROM-like device. Essentially what this table is indicating is that the DC-SCM contains identification that can be gathered externally from management interfaces that the BMC exposes.

5. HPM FPGA Interoperability Requirements

This section specifies common HPM FPGA requirements for interoperability.

This section does not teach specific implementations but rather outlines what is required vs. recommended and optional.

This section will refer to external design specifications with HPM FPGA specification being primarily referenced.

6. Component FRU Requirements

This section specifies requirements for interoperability regarding the component FRU contents.

This section does not teach specific implementations but rather outlines what is required vs. recommended and optional.

This section will refer to external design specifications for reference.

7. Discovery Process Requirements

This section specifies requirements for interoperability regarding the discovery process.

This section does not teach specific implementations but rather outlines what is required vs. recommended and optional.

This section will refer to external design specifications for reference.

8. Common Boot Sequence and Process Interoperability Requirements

This section specifies requirements for interoperability regarding the common **Boot Sequence and Process** for MHS platforms. This section does not teach specific implementations but rather outlines what is required vs. recommended and optional.

9. Runtime System Management and Sensor Requirements

This section specifies common requirements for runtime system management expected by the SCM of the HPM and other system components.

This section does not teach specific implementations but rather outlines what is required vs. recommended and optional.

10. Security Requirements

This section specifies common requirements for system security expected by the SCM of the HPM and other system components.

This section does not teach specific implementations but rather outlines what is required vs. recommended and optional.

This section will refer to external design specifications with MHS System Security Design Specification being primarily referenced.

11. Non-Functional Requirements

This section specifies common non-functional requirements for the system.

This section does not teach specific implementations but rather outlines what is required vs. recommended and optional.

12. Specification Compliance Checklist

The following sections contain compliance checklists for each MHS component and functional area.

The following table is intended to summarize the list of attributes and requirements for a design to be MHS M-PIC Base Specification compliant. Compliance requirements throughout this document can be identified by underlined text including the word “shall”.

#	Technical Specification	Document Reference
1		
2		

13. Trademarks

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14. Future Material

Note: This section is for reference only and intends to move into the main document body when further refined by the workgroups after Release 1.0. The implementation is highly subject to change.